

The compact-tail kernel/Loewner method at $L = 0.8$

a sealed method no-go beyond the prime-free zone

PRIME.RDAGGER.KERNEL_LOEWNER.02 — round 496

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Abstract

The round-492 three-gate contract is tested at $L = 0.8$, where the prime powers 2, 3, 4 enter. Gate G1 passes: the x -space identity, its factor-two prime normalization, the zero-extension convention, and the r471/r477 calibration stops agree. Gates G2 and G3 give an honest method no-go. The best tested booked cutoff already has negative 101-section margin, while its full three-piece Hilbert–Schmidt charge is 27.3672. More fundamentally, the prime translations are bounded but noncompact; their measured Legendre off-block does not vanish. Therefore the compact-kernel completion used at $L = 0.3$ cannot treat the prime section as the whole operator. This excludes one method only. It proves neither negativity nor failure of Weil positivity at $L = 0.8$, and it is not a claim for or against RH.

Verdict. NO_GO(kernel-Loewner-compact-tail@L=0.8).

Probe SPEC_SHA: 63f590d43c9ed63e8419c99ddd6b9110e58a7f419d6445de71932aa4575cba97.

Claim boundary. This note certifies a failure of the prescribed compact-tail proof budget, not a counterexample and not an anti-positivity theorem. The positive r471 and r477 finite-subspace calibrations remain valid. No Galerkin value is promoted to an operator value. No RH claim.

1 Exact prime normalization and the first balance

For real h supported in $[-L, L]$, extend h by zero to $L^2(\mathbb{R})$ and put $g = h * \tilde{h}$. The r471 classical normalization is

$$P(h) = 2 \sum_{n \in \{2,3,4\}} \frac{\Lambda(n)}{\sqrt{n}} g(\log n).$$

Thus, with $a_n = \Lambda(n)/\sqrt{n}$,

$$-P(h) = -2 \sum_n a_n g(0) + 2 \sum_n a_n \langle h, (I - \operatorname{Re} \tau_{\log n}) h \rangle.$$

The second sum is positive. The first is the full prime $g(0)$ load. The factor two is essential: omitting it misses both r471 and r477.

At $L = 0.8$ the interval computation gives

$$\begin{aligned} c_L &= 3.5598735700355446228812256099, \\ (a_2, a_3, a_4) &= (0.4901290717342736, 0.6342841005975640, 0.3465735902799727), \\ 2 \sum_n a_n &= 2.9419735252236204555039093902. \end{aligned}$$

For the r494 cutoff $[0.01, 0.03]$, $\kappa_w = 5.065476699202523$ and the scalar balance $\kappa_w - c_L - 2 \sum a_n = -1.4363703961$ is already negative. Only substantially harder cutoffs make that scalar line positive.

2 Gate G1: identity passes

Five compactly supported polynomial test functions were evaluated independently in the x -space form and in the defining frequency form at `mp.dps=50`. The frequency multiplier included

$$-2 \sum_{n \in \{2,3,4\}} a_n \cos(t \log n).$$

The maximum normalized discrepancy was $5.499 \cdot 10^{-14}$. After this gate, no frequency quantity was used again.

The support convention was tested by the r495-A2 rational staircase battery: all six identities

$$2\{g(0) - g(a)\} = \|h - \tau_a h\|_{L^2(\mathbb{R})}^2$$

held exactly over $\mathbb{Q}$, including the boundary strips omitted by an interval-only norm.

Calibration stop	Enclosure/value	Result
r471, $L = 0.8$, $n = 24$	hint $2.64507571918113 \cdot 10^{-4}$, $\mu_{\text{lo}} = 1.32253673213734 \cdot 10^{-4}$	pass
r477 constant mode	$r(1) \in [0.0802740548742, 0.0802740551870]$	pass
r477 (1, first Dirichlet)	Schur $\in [0.0741382774823, 0.0750735086403]$	pass

3 Gate G2: the booked cutoff budget fails

Let w be zero on $[0, x_0]$, one on $[x_1, 1.6]$, and the quintic C^2 smoothstep between them. Every row below includes c_L , the full factor-two prime $g(0)$ mass, and the omitted archimedean integral through the Loewner minorant.

x_0	x_1	κ_w	scalar balance	$\lambda_{\min}(P_{101}Q_wP_{101})$
0.001	0.003	7.3770530633	$+8.752060 \cdot 10^{-1}$	$-1.6106733 \cdot 10^{-4}$
0.002	0.006	6.6829061429	$+1.810590 \cdot 10^{-1}$	$-8.1766226 \cdot 10^{-4}$
0.003	0.010	6.1992956339	$-3.025515 \cdot 10^{-1}$	$-2.2841017 \cdot 10^{-3}$
0.010	0.030	5.0654766992	$-1.436370 \cdot 10^0$	$-3.0322577 \cdot 10^{-2}$

The best tested row already has negative finite-section margin. By Cauchy interlacing, every containing Legendre section through $n = 800$ has minimum eigenvalue no larger than this value. Hence there is no positive finite margin to which a remainder could be compared.

4 Gate G3: compact-tail completion fails

For

$$T_w(y, z) = \frac{1}{2} w(|y - z|) \frac{e^{|y-z|/2}}{\sinh |y - z|},$$

the exact Hilbert–Schmidt norm is

$$\|T_w\|_{\text{HS}}^2 = \frac{1}{2} \int_0^{1.6} (1.6 - x) \left(w(x) \frac{e^{x/2}}{\sinh x} \right)^2 dx.$$

For the best G2 row,

$$\|T_w\|_{\text{HS}} = 19.217696275, \quad r_{101} = 9.122410180, \quad 3r_{101} = 27.367230540.$$

The full $r + 2r$ charge is used; neither $r/3$ nor the central block alone is admitted.

There is a second, structural obstruction. A compressed translation has a delta-line kernel. It is bounded, but it is not compact or Hilbert–Schmidt. Exact Gauss integration of the prime translation matrix gives

$$\left\| P_{[101,301)} \left(2 \sum_n a_n \operatorname{Re} R_{\log n} \right) P_{101} \right\| = 0.939496613,$$

with Frobenius norm 3.807582717. Consequently its finite Legendre section cannot be substituted for the operator and cannot be absorbed into the r494 Hilbert–Schmidt tail. Doing so would be precisely the forbidden Galerkin-as-operator pad.

Proposition 1 (Round-496 method no-go). *Under the round-492/r494 three-gate contract, the compact-kernel finite-section method does not prove*

$$\lambda_*(0.8) \geq c$$

for any $c > 0$. Gate G1 passes, but G2 has no positive certified finite margin and G3 cannot supply a remainder below one third of that margin at any containing section with $n \leq 800$.

5 Kill and pad audit

Binding item	R496 disposition
Galerkin as operator	blocked; section values are obstruction diagnostics only
Unbooked $[0, x_0]$	blocked; every row uses its actual κ_w
HS without full $3\times$ charge	blocked; $r + 2r = 3r$ is reported
Frequency re-entry	blocked after the five-function G1 seal
Prime support convention	r495-A2 zero-extension battery passes 6/6 exactly
Prime section treated as finite rank	rejected; nonvanishing off-block is measured explicitly

6 What remains open

The r471 grid certificate and r477 two-dimensional certificate are finite-subspace positives, not lower bounds for $\lambda_*(0.8)$. This round does not decide whether another representation can prove full-class positivity at $L = 0.8$. Any successor would need a noncompact-translation argument, rather than the compact Hilbert–Schmidt completion that closed r494. Per the r492 budget, this lane stops here.